Machine Learning Model Regularization

Decision Tree vs. Bagging Ensemble

Part 4: Comparison and Contrast

Course: Machine Learning

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Abstract

This report examines the impact of ensemble-based regularization on a high-variance machine learning model. A fully grown Decision Tree classifier is compared against a Bagging ensemble trained on the same dataset. Experimental results show that Bagging substantially improves generalization performance by reducing variance while preserving low bias. The findings demonstrate that Bagging functions as an effective implicit regularization technique for complex models.

Introduction

Overfitting is a fundamental challenge in machine learning, particularly for highly expressive models such as Decision Trees. When trained without constraints, Decision Trees can perfectly fit training data, including noise, resulting in poor generalization to unseen data.

Regularization techniques aim to address this problem. Explicit regularization methods directly constrain model complexity, while ensemble methods such as Bagging provide implicit regularization by aggregating multiple high-variance learners. This report compares an overfit Decision Tree with a Bagging ensemble to analyze how ensemble averaging influences generalization.

Experimental Setup and Results

Two models were trained on the same classification task:

- An unconstrained Decision Tree
- A Bagging ensemble consisting of 200 bootstrap-trained Decision Trees

Performance was evaluated using training accuracy, test accuracy, ROC AUC, weighted F1-score, and the overfitting gap defined as the difference between training and test accuracy.

Table 1: Performance Comparison

Metric	**Decision Tree**	**Bagging**	**Change**
Training Accuracy	100.00%	100.00%	—
Test Accuracy	92.31%	95.10%	+2.79%
ROC AUC	0.9234	0.9941	+0.0707
Weighted F1	0.9235	0.9510	+0.0275
Overfit Gap	7.69%	4.90%	-2.79%

The Bagging ensemble outperforms the single Decision Tree across all generalization metrics, with a clear reduction in overfitting.

Analysis and Discussion

The unconstrained Decision Tree exhibits classic high-variance behavior. Perfect training accuracy indicates memorization of the training data, while the large overfitting gap demonstrates limited generalization capability.

In contrast, the Bagging ensemble significantly improves test performance while maintaining perfect training accuracy. The overfitting gap is reduced by approximately 36%, confirming that Bagging regularizes the model without reducing its expressive power.

Theoretical Explanation

Bagging reduces variance by averaging predictions from multiple models trained on bootstrap samples. Let each base learner have variance σ^2 and pairwise correlation ρ. The variance of an ensemble of B models is given by:

$$\text{Var(Ensemble)} = \rho\sigma^2 + \frac{(1-\rho)\sigma^2}{B} \tag{1}$$

As B increases, the second term approaches zero, significantly reducing variance as long as $\rho < 1$. In the special case of independent learners ($\rho = 0$), variance decreases proportionally to $1/B$.

Since Bagging does not alter the hypothesis space of the base learner, it does not increase bias. Thus, the ensemble preserves the low bias of deep Decision Trees while reducing variance.

Conclusion

This report demonstrates that Bagging is an effective implicit regularization technique for high-variance models. By aggregating multiple overfitting Decision Trees, the ensemble achieves improved generalization, higher discrimination performance, and a substantially reduced overfitting gap.

The results highlight the theoretical and practical importance of ensemble methods as a regularization strategy in machine learning.